

Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation
Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane

★ **CORE**

Let

$$S = \{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{k}^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x - 1, y) \cap (x, y - 1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = (xy, x(x - 1), y(y - 1)).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\mathbb{k}[x, y]/\mathbf{I}(S) \simeq \mathbb{k}^3.$$

Exercise 2. Two axes and their coordinate rings

★ **CORE**

In $\mathbb{k}^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \quad X \cap Y = \mathbf{V}(x, y).$$

(ii) Compute the coordinate rings of X , Y , $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\mathbb{k}[x, y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp

★★ **CORE**

Consider

$$P = \mathbf{V}(y - x^2), \quad C = \mathbf{V}(y^2 - x^3) \subset \mathbb{k}^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{k} \rightarrow P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{k} \rightarrow C$ which is bijective on the underlying sets.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Automorphisms of the affine and rational line

★★ **CHOICE**

(i) Show that morphisms of affine algebraic sets $\mathbb{k} \rightarrow \mathbb{k}$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

(ii) Prove that such a morphism is an automorphism exactly when $f(t) = at + b$ with $a \in \mathbb{k}^\times$. Deduce that

$$\mathrm{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}[t]) \simeq \mathbb{k} \rtimes \mathbb{k}^\times.$$

(iii) After projective space has been introduced, show that

$$t \mapsto \frac{at + b}{ct + d}, \quad ad - bc \neq 0,$$

defines a $\mathbb{k}$ -automorphism of $\mathbb{k}(t)$ and that scalar multiples of the matrix give the same automorphism. Check the order under composition. Using the standard fact $[\mathbb{k}(t) : \mathbb{k}(r)] = \deg(r)$ for a non-constant rational function r , prove that

$$\mathrm{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}(t)) \simeq \mathrm{PGL}_2(\mathbb{k}).$$

Which subgroup preserves the subring $\mathbb{k}[t]$?

Exercise 5. Equations, zero sets and ideals of subsets★★ **CORE**Work over $\mathbb{R}$.

- (i) Show that, for every
- $m \geq 2$
- , the three distinct ideals

$$(x, y), \quad (x^2, xy, y^3), \quad (x, y)^m$$

define the same subset $\{(0, 0)\} \subset \mathbb{R}^2$. Determine the ideal of polynomials vanishing on this subset.

- (ii) Let
- $C = \{\pm 1\}^n \subset \mathbb{R}^n$
- be the set of vertices of the
- n
- dimensional cube. Prove that

$$\mathbf{I}(C) = (x_1^2 - 1, \dots, x_n^2 - 1).$$

- (iii) Regard
- $\mathbb{Z}$
- as a subset of
- $\mathbb{R}$
- . Compute
- $\mathbf{I}(\mathbb{Z}) \subset \mathbb{R}[t]$
- . Deduce that
- $\mathbb{Z}$
- is not an algebraic subset of the affine line over
- $\mathbb{R}$
- .

Exercise 6. Polynomial systems versus linear systems★ **CORE**Fix $n, d \geq 1$.

- (i) Recall why an invertible linear system of
- n
- equations in
- n
- unknowns has exactly one solution.
-
- (ii) Over
- $\mathbb{C}$
- , determine all solutions of

$$x_1^d = 1, \quad x_2^d = 1, \quad \dots, \quad x_n^d = 1.$$

Show that there are exactly d^n solutions.

- (iii) Write a system of
- n
- polynomial equations in
- n
- variables having no solution over
- $\mathbb{R}$
- .
-
- (iv) Explain precisely which expectations imported from linear algebra have failed in (ii) and (iii).

Thursday afternoon — material available: Lectures 1–4**Exercise 7. Radicals and reduced rings**★ **CORE**

Compute

$$\sqrt{(x^2, y^3)}, \quad \sqrt{(x^2 y^3)}$$

in $\mathbb{k}[x, y]$. Decide which of the rings

$$\mathbb{k}[x]/(x^2), \quad \mathbb{k}[x, y]/(xy), \quad \mathbb{k}[x, y]/(x^2, xy)$$

are reduced.

Exercise 8. Points at infinity★ **CORE**In the affine chart $x_0 \neq 0$ of $\mathbf{P}_{\mathbb{k}}^2$, write $x = x_1/x_0$ and $y = x_2/x_0$.

- (i) Homogenise the parabola
- $y = x^2$
- and determine its points at infinity.
-
- (ii) Homogenise the hyperbola
- $xy = 1$
- and determine its points at infinity.
-
- (iii) Explain geometrically why the answers are different.

Exercise 9. The line through two points of projective space★ **CHOICE**Let $p = [a_0 : \dots : a_n]$ and $q = [b_0 : \dots : b_n]$ be distinct points of $\mathbf{P}_{\mathbb{k}}^n$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.**Exercise 10. Points in $\mathbf{P}^2$ from minors**★★ **CHOICE**Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbf{P}^2$. For four lines $\ell_1, \dots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Exercise 11. Equations of the Segre and Veronese images

*** OPTIONAL

a) For

$$([s_0 : s_1], [t_0 : t_1]) \mapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbf{P}^3$.

b) For the quadratic Veronese map $\mathbf{P}^2 \rightarrow \mathbf{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Exercise 12. Localising the coordinate axes

★ CORE

Let

$$A = \mathbb{k}[x, y]/(xy).$$

- (i) Compute A_x and A_y .
- (ii) Compute $A_{(x)}$ and $A_{(y)}$.
- (iii) Explain what these localisations retain and what they discard geometrically.

Exercise 13. Localisation with a nilpotent

★ CHOICE

Let $A = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \quad A_{(\varepsilon)}, \quad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 14. Two finite module structures

★★ CHOICE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\mathbb{k}[x]$ and free of rank 1 over $\mathbb{k}[y]$. Compare geometrically the corresponding morphisms to the affine line.

Exercise 15. Noether normalisation for the hyperbola

*** CHOICE

Let

$$B = \mathbb{k}[x, y]/(xy - 1), \quad u = x + y \in B.$$

Show that B is finite over $\mathbb{k}[u]$. Deduce that the corresponding morphism to the affine line $\mathbb{k}$ has fibres with at most two points.

After Friday's lecture or homework

Exercise 16. A presheaf which is not a sheaf

★ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \emptyset, \quad \mathcal{F}(\emptyset) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 17. Continuous and holomorphic functions

★★ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 18. $\mathcal{O}(d)$ on the projective line

*** OPTIONAL

Identify $\mathbf{P}_{\mathbb{C}}^1$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞ . Compute its global sections.